

Question number	Answer	Additional guidance	Mark
7(a)	One from • high BMI / smoking / alcohol intake / high salt intake / high cholesterol intake And one from • age / sex / gender	Accept: obesity / overweight / high waist to hip ratio high level of {fat / sugar} in diet type 2 diabetes high stress levels air pollution	(1)

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7(b)	A description that includes two of the following points: • increase in intensity of exercise decreases the risk of death from heart disease (1) • increase in the energy needed for exercise decreases the risk of death from heart disease (1) • for the same energy expenditure, {vigorous exercise / moderate exercise} decreases risk of death from CVD more than light exercise (1)	Accept converse throughout Accept if you expend 4000kJ per week, intensity of exercise makes no difference to risk of death from CVD	(2)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An explanation that includes the following points: • because antioxidants reduce free radicals (1) • free radicals cause {cell damage /tissue damage / oxidative stress / damage to endothelial lining} (1) • (antioxidants) reduce {plaque / atheroma} formation (1)	Accept neutralise / stabilise / donate electrons Accept antioxidants {prevent cell damage / tissue damage / reduce oxidative stress / prevent damage to endothelial lining} Accept reduces {cholesterol build up / blood clot formation / atherosclerosis}	(3)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that includes three of the following points: • use (a large number of) {healthy individuals / individuals with no known heart condition} (1) • who have similar (lifestyle and non-lifestyle) risk factors (1) • compare group given antioxidants to a group using other preventative treatments (1) • monitor the incidence of heart disease over a (long) period of time (1)	Accept control for named factor Accept give one group antioxidants and the other group {a placebo / no antioxidants} Accept at least 6 months if time is given	(3)

Question number	Answer						
*7(d)	Indicative content: <table border="1" data-bbox="454 256 2063 986"> <tr> <td data-bbox="454 256 869 416">Outline of events (O points)</td><td data-bbox="869 256 2063 416"> 1. less blood reaches the heart muscle cells, 2. muscle cells are not supplied with enough oxygen 3. muscle cells are not supplied with enough glucose 4. aerobic respiration decreases </td></tr> <tr> <td data-bbox="454 416 869 699">Contraction stops (C points)</td><td data-bbox="869 416 2063 699"> 5. anaerobic respiration produces lactic acid 6. lactic acid lowers pH and denatures enzymes 7. glycogen used as respiratory substrate 8. anaerobic respiration produces much less ATP / energy 9. therefore less / no blood is pumped by the heart 10. reference to effect on other organs </td></tr> <tr> <td data-bbox="454 699 869 986">Use of graph / stem info (G points)</td><td data-bbox="869 699 2063 986"> 11. energy released from heart cells decreases with time after blockage 12. reference to data from graph (not mp13 or 15) 13. after 8 minutes the energy released is 52 - 54 a.u. 14. after 8 minutes there is not enough energy for contraction 15. after 20 minutes the energy released is 23 - 24 a.u. 16. which is too low to maintain cell viability / for cells to survive 17. cells die as not enough ATP for vital processes eg active transport </td></tr> </table> Level 1: up to two points from anywhere; one point for 1 mark and two points for 2 marks Level 2: up to two points from each of two categories; 3 points for 3 marks, 4 points for 4 marks Must achieve 4 marks to progress to level 3 Level 3: Level 2 plus two points from the third category; five points for 5 marks and six points for 6 marks	Outline of events (O points)	1. less blood reaches the heart muscle cells, 2. muscle cells are not supplied with enough oxygen 3. muscle cells are not supplied with enough glucose 4. aerobic respiration decreases	Contraction stops (C points)	5. anaerobic respiration produces lactic acid 6. lactic acid lowers pH and denatures enzymes 7. glycogen used as respiratory substrate 8. anaerobic respiration produces much less ATP / energy 9. therefore less / no blood is pumped by the heart 10. reference to effect on other organs	Use of graph / stem info (G points)	11. energy released from heart cells decreases with time after blockage 12. reference to data from graph (not mp13 or 15) 13. after 8 minutes the energy released is 52 - 54 a.u. 14. after 8 minutes there is not enough energy for contraction 15. after 20 minutes the energy released is 23 - 24 a.u. 16. which is too low to maintain cell viability / for cells to survive 17. cells die as not enough ATP for vital processes eg active transport
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